

Shubham Shukla

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Profile Summary

IT undergraduate with hands-on experience building and securing applications for a Government of India law-enforcement unit (I4C, Ministry of Home Affairs). Skilled in designing secure authentication systems, diagnosing critical API vulnerabilities, and shipping production-grade software under tight timelines. Strong applied background in full-stack development, edge AI systems, and application security.

Education

University School of Information, Communication & Technology (USICT), GGSIPU Aug 2024 – Jul 2028
B.Tech in Information Technology
CGPA (till 3rd Semester): 8.19
Sardar Patel Public Sr. Sec. School, Karawal Nagar, Delhi Apr 2021 – Apr 2023
Higher Secondary Education – 76.6%

Experience

App Development Intern (Onsite) Jan 2026 – Mar 2026
IFSO Special Cell (I4C, Ministry of Home Affairs, Government of India)

- Engineered a production-grade Visitor Management System for a national cyber-crime law-enforcement unit, implementing JWT-based authentication and role-based access control (RBAC) to secure visitor session data – shipped as a live APK in active use by the Ministry of Home Affairs.
- Conducted security-focused debugging of the integration layer – independently diagnosed and resolved critical JSON parsing failures and API vulnerabilities that were blocking secure deployment.
- Collaborated with ministry stakeholders to deliver a secure, offline-resilient mobile application under strict government compliance timelines.

Projects

Secure Visitor Management System – Government Access Control Platform Jan 2026 – Mar 2026
React Native, Node.js, JWT, RESTful APIs

- Designed an end-to-end identity and access management flow – visitor registration, approval gating, and session expiry – enforcing JWT-based authentication and RBAC across all endpoints to prevent unauthorized access.
- Hardened API integration points against malformed-payload failures after identifying a critical JSON parsing vulnerability in production; rebuilt input validation to close the gap before deployment.
- Built audit-friendly session logs for visitor entry/exit, supporting traceability requirements expected in sensitive government facilities.

NavDrishti – Real-Time Edge AI Monitoring System for Assistive Navigation | *Python, Raspberry Pi, YOLOv8, FastAPI, ZeroMQ, WebSockets*

- Built a real-time edge monitoring pipeline on Raspberry Pi 4 – continuous detection → contextual hazard/threat analysis → instant alerting – achieving sub-second latency under hardware constraints.
- Designed a distributed, fault-tolerant multi-worker inference system (ZeroMQ + FastAPI + WebSockets) for resilient parallel processing and live session monitoring.
- Implemented secure WebSocket channels between edge device and dashboard, with structured error handling to prevent data loss under unstable network conditions.

RAG-based Intelligent Document Assistant – Secure Document Intelligence Pipeline | *Python, LangChain, FAISS, Gemini, FastAPI, Streamlit*

- Architected an end-to-end RAG pipeline for automated analysis of sensitive document sets – ingestion → chunking → vector retrieval → verified answer synthesis – with sub-second query latency on 100-page documents.
- Optimized chunk-size and similarity re-ranking thresholds to reduce hallucinated/irrelevant responses, improving retrieval accuracy and answer trustworthiness.
- Deployed via FastAPI + Streamlit with isolated processing per document session, minimizing cross-session data exposure.

Achievements

Qualified – Smart India Hackathon (SIH) Internal Round, GGSIPU 2025

- Selected through competitive internal screening; awarded Certificate of Appreciation for problem-solving and solution design.

Skills

Security & Systems: JWT Authentication, RBAC, RESTful API Security, Session Management, Vulnerability Diagnosis & Resolution, Secure Deployment, Input Validation

Programming & Frameworks: Python, C/C++, Java, JavaScript, FastAPI, Flask, React, React Native, Node.js

AI/ML & Tools: NLP, Computer Vision, RAG, Object Detection, Edge AI, Git, GitHub, Firebase, MySQL, ZeroMQ